

Theoretical Foundations and Information-Theoretic Bounds on Neural Scaling Laws: A Mathematical Monograph

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Abstract

Empirical scaling laws in deep learning demonstrate that cross-entropy loss decreases as a power law with respect to parameter count N , dataset size D , and compute budget C , following the parametric form $L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}$. While originally discovered as empirical regularities, a rigorous theoretical foundation has emerged connecting neural scaling laws to four major mathematical paradigms: **(i) Information Theory and Rate-Distortion Theory**, which provides the correct Shannon-theoretic analogue via reverse water-filling on heavy-tailed spectra and Yang-Barron metric entropy bounds; **(ii) Statistical Language Physics and Long-Range Correlations**, which analytically derives the data scaling exponent $\alpha_D = \gamma/(2\beta)$ from Hilberg’s law and token covariance decay; **(iii) High-Dimensional Differential Geometry and Approximation Theory**, which maps parameter exponents $\alpha_N \approx 4/d$ to the intrinsic dimension of the data manifold and establishes fundamental continuous nonlinear n -width impossibility bounds; and **(iv) Statistical Physics and Spectral Decomposition**, which derives generalization curves $\mathcal{E}(D) \sim D^{-\frac{a+b-1}{a}}$ via Neural Tangent Kernel (NTK) spectra and random matrix theory. This monograph provides a unified, self-contained, and rigorous survey of the mathematical machinery underpinning neural scaling laws, contrasts channel coding error exponents with rate-distortion functions, catalogs fundamental impossibility bounds, analyzes discrete quantization models, and highlights open challenges regarding the “honest gap” between theoretical parameterizations and empirical observation.

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1 Introduction and Empirical Landscape

Over the past decade, empirical deep learning has transitioned from heuristic architecture search to predictable scaling engineering. Landmark empirical studies by Kaplan et al. [1] and Hoffmann et al. (Chinchilla) [2] demonstrated that the cross-entropy loss L of autoregressive language models scales as a smooth power-law function over several orders of magnitude in parameter count N , training token count D , and total floating-point operations $C \approx 6ND$:

$$L(N, D) = E + \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D}, \quad (1)$$

where E denotes the irreducible entropy floor (the Bayes risk of the underlying natural language data generation process), N_c, D_c are normalization constants, and $\alpha_N, \alpha_D > 0$ are characteristic scaling exponents. For standard autoregressive Transformers on natural English text, empirical measurements typically find:

$$\alpha_N \approx 0.076 \text{ to } 0.34, \quad \alpha_D \approx 0.095 \text{ to } 0.28, \quad E \approx 1.5 \text{ to } 2.1 \text{ nats/token}. \quad (2)$$

Despite the empirical ubiquity of power laws across modalities—including text, vision, audio, and code—fundamental theoretical questions remain:

- (i) **Why power laws?** Why do learning curves exhibit polynomial decay $\mathcal{O}(N^{-\alpha})$ rather than exponential decay $\mathcal{O}(e^{-cN})$ or logarithmic stagnation $\mathcal{O}(1/\log N)$?
- (ii) **What sets the exponents?** Are α_N and α_D universal constants of neural architectures, or are they fingerprints of the geometry, spectral decay, and correlation structure of the data distribution?
- (iii) **What is the Shannon connection?** How does neural scaling relate to classical information theory, channel coding, rate–distortion theory, and the Kolmogorov ε -entropy?
- (iv) **What are the fundamental limits?** What mathematical lower bounds (from approximation theory, algorithmic information theory, and computational complexity) govern any continuous learning algorithm?

This monograph provides a rigorous, self-contained mathematical synthesis of the theoretical bounds and frameworks that answer these questions.

2 Information-Theoretic Foundations: Shannon Limits and Rate–Distortion

2.1 Channel Coding Reliability Exponent vs. Rate–Distortion Theory

A frequent conceptual intuition posits that neural scaling mirrors the classical *channel coding reliability exponent*. We formalize this distinction and show why the appropriate classical analogue is **rate–distortion theory** rather than channel error exponents.

Definition 2.1 (Channel Reliability Error Exponent). In Shannon channel coding, for a discrete memoryless channel with capacity C and a transmission rate $R < C$, the optimal average probability of block decoding error P_e decays *exponentially* with blocklength n :

$$P_e(n, R) \asymp \exp(-nE_r(R)), \quad (3)$$

where $E_r(R) = \max_{P_X} \max_{\rho \in [0,1]} [E_0(\rho, P_X) - \rho R] > 0$ is Gallager’s random coding error exponent.

In contrast, empirical neural scaling curves obey polynomial power laws:

$$L(N) - E \propto N^{-\alpha_N}. \quad (4)$$

While taking the logarithm of both sides yields linear relationships in log-space ($\log(L - E) = \text{const} - \alpha_N \log N$ versus $\log P_e \sim -nE(R)$), the mathematical mechanisms are fundamentally distinct:

- Channel coding error exponents govern the tail probability of typical sets in product spaces under independent channel noise as blocklength $n \rightarrow \infty$.
- Neural scaling measures the *excess distortion* (Kullback–Leibler divergence or mean squared error) in approximating a continuous or combinatorial high-dimensional source using a compressed representation of size N (or $R \approx Nb$ bits of precision).

Theorem 2.1 (Distortion–Rate Function and Spectral Reverse Water-Filling). Let $X \sim \mathcal{N}(0, \Sigma)$ be an infinite-dimensional Gaussian random vector in a Hilbert space with covariance eigenvalues decaying as a power law:

$$\lambda_k = c \cdot k^{-(1+\beta)}, \quad \beta > 0, \quad k \in \mathbb{N}^+. \quad (5)$$

Under mean squared error distortion $D = \mathbb{E}[\|X - \hat{X}\|^2]$, the distortion–rate function $R(D)$ and its inverse, the rate–distortion function $D(R)$, are determined parametrically via reverse water-filling:

$$D(\theta) = \sum_{k=1}^{\infty} \min(\theta, \lambda_k), \quad R(\theta) = \sum_{k=1}^{\infty} \frac{1}{2} \log_2 \max\left(1, \frac{\lambda_k}{\theta}\right), \quad (6)$$

where $\theta > 0$ is the water-filling threshold. In the asymptotic regime as $R \rightarrow \infty$ ($\theta \rightarrow 0$):

$$\boxed{D(R) \asymp R^{-\beta}}. \quad (7)$$

Proof. Let $K(\theta) = \max\{k : \lambda_k \geq \theta\}$. From $\lambda_k = ck^{-(1+\beta)} = \theta$, we have $K(\theta) = (c/\theta)^{\frac{1}{1+\beta}}$. The description rate is:

$$\begin{aligned} R(\theta) &= \sum_{k=1}^{K(\theta)} \frac{1}{2} \log_2 \left(\frac{ck^{-(1+\beta)}}{\theta} \right) \approx \frac{1}{2 \ln 2} \int_0^{K(\theta)} \left[\ln \left(\frac{c}{\theta} \right) - (1+\beta) \ln x \right] dx \\ &= \frac{1}{2 \ln 2} \left[K(\theta) \ln \left(\frac{c}{\theta} \right) - (1+\beta) (K(\theta) \ln K(\theta) - K(\theta)) \right] \\ &= \frac{1+\beta}{2 \ln 2} K(\theta) = \frac{1+\beta}{2 \ln 2} \left(\frac{c}{\theta} \right)^{\frac{1}{1+\beta}} \asymp \theta^{-\frac{1}{1+\beta}}. \end{aligned} \quad (8)$$

Inverting this relationship gives $\theta(R) \asymp R^{-(1+\beta)}$. The total distortion $D(\theta)$ decomposes into two terms:

$$\begin{aligned} D(\theta) &= \sum_{k=1}^{K(\theta)} \theta + \sum_{k=K(\theta)+1}^{\infty} \lambda_k = \theta K(\theta) + \sum_{k=K(\theta)+1}^{\infty} ck^{-(1+\beta)} \\ &\approx \theta \left(\frac{c}{\theta} \right)^{\frac{1}{1+\beta}} + \int_{K(\theta)}^{\infty} cx^{-(1+\beta)} dx = c^{\frac{1}{1+\beta}} \theta^{\frac{\beta}{1+\beta}} + \frac{c}{\beta} K(\theta)^{-\beta} \\ &= c^{\frac{1}{1+\beta}} \theta^{\frac{\beta}{1+\beta}} + \frac{c}{\beta} \left[\left(\frac{c}{\theta} \right)^{\frac{1}{1+\beta}} \right]^{-\beta} = \left(1 + \frac{1}{\beta} \right) c^{\frac{1}{1+\beta}} \theta^{\frac{\beta}{1+\beta}}. \end{aligned} \quad (9)$$

Substituting $\theta(R) \asymp R^{-(1+\beta)}$ into $D(\theta)$:

$$D(R) \asymp \left(R^{-(1+\beta)} \right)^{\frac{\beta}{1+\beta}} = R^{-\beta}. \quad (10)$$

□

Corollary 2.2 (Finite Dimension vs. Infinite-Dimensional Heavy Tail). If the source is strictly d -dimensional ($\lambda_k = 0$ for $k > d$), reverse water-filling yields exponential decay:

$$D(R) \asymp 2^{-2R/d} = \exp\left(-\frac{2 \ln 2}{d} R\right). \quad (11)$$

Core Theoretical Insight: The Fingerprint of Infinite Dimensions

A power-law scaling curve $L(N) \sim N^{-\alpha}$ is the **direct mathematical fingerprint of an effectively infinite-dimensional, heavy-tailed source eigenvalue spectrum** ($\lambda_k \sim k^{-(1+\beta)}$). If data lived in a finite-dimensional Euclidean subspace with bounded spectrum, rate-distortion would mandate exponential decay $D(R) \sim 2^{-2R/d}$. Thus, neural scaling laws are a direct consequence of the infinite-dimensional, self-similar spectral structure of natural data.

2.2 The Shannon Entropy Floor E as an Irreducible Bayes Lower Bound

In autoregressive next-token prediction over an alphabet $\mathcal{V}$, sequence $X = (X_1, X_2, \dots)$ is modeled as a stationary ergodic stochastic process. The cross-entropy loss with context length T for model $P_\theta(X_{t+1} \mid X_1^t)$ satisfies:

$$\mathcal{L}(P_\theta) = - \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{X \sim P^*} [\log P_\theta(X_{t+1} \mid X_1^t)] = H(\mathcal{X}) + D_{\text{KL}}(P^* \parallel P_\theta), \quad (12)$$

where $H(\mathcal{X})$ is the *entropy rate* of natural language:

$$H(\mathcal{X}) = \lim_{t \rightarrow \infty} H(X_{t+1} \mid X_1, \dots, X_t) = \lim_{t \rightarrow \infty} \frac{1}{t} H(X_1, \dots, X_t). \quad (13)$$

By the non-negativity of Kullback–Leibler divergence ($D_{\text{KL}}(P^* \parallel P_\theta) \geq 0$), we have the absolute lower bound:

$$L(N, D) \geq E = H(\mathcal{X}) \quad \forall N, D \in \mathbb{N}. \quad (14)$$

Shannon’s 1951 human prediction experiments [3] estimated $H(\mathcal{X}) \approx 0.6$ to 1.3 bits/character. Modern frontier language models achieve cross-entropy losses between 1.0 and 1.5 nats/token (roughly 0.6–0.9 bits/char), closely approaching this fundamental Shannon limit.

2.3 Yang–Barron Information-Theoretic Metric Entropy Bounds

Yang and Barron (1999) [4] developed the definitive minimax lower and upper bounds for density estimation and statistical learning using the Kolmogorov ε -entropy.

Definition 2.2 (Kolmogorov ε -Entropy). Let $(\mathcal{F}, \rho)$ be a class of target functions or probability densities equipped with metric ρ (e.g., Hellinger distance d_H or total variation d_{TV}). The ε -covering number $N(\varepsilon, \mathcal{F}, \rho)$ is the minimal number of balls of radius ε needed to cover $\mathcal{F}$. The Kolmogorov metric entropy is:

$$H(\varepsilon; \mathcal{F}) = \log_2 N(\varepsilon, \mathcal{F}, \rho). \quad (15)$$

Theorem 2.3 (Yang–Barron Minimax Theorem [4]). Let $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$ be a family of probability distributions on $\mathcal{X}$. Suppose that for small $\varepsilon > 0$, the metric entropy under Hellinger distance satisfies:

$$H(\varepsilon; \mathcal{P}) \asymp \varepsilon^{-r} \quad \text{for some } r > 0. \quad (16)$$

Then the minimax estimation risk over n independent and identically distributed samples satisfies:

$$\inf_{\hat{p}_n} \sup_{p^* \in \mathcal{P}} \mathbb{E}_{p^*} [D_{\text{KL}}(p^* \parallel \hat{p}_n)] \asymp \varepsilon_n^2, \quad (17)$$

where the minimax optimal rate ε_n is the unique solution to the Yang–Barron balance equation:

$$\boxed{H(\varepsilon_n; \mathcal{P}) \asymp n \varepsilon_n^2}. \quad (18)$$

Corollary 2.4 (Sobolev Smoothness in Dimension d). For target functions in the Sobolev space $W^s(\mathbb{R}^d)$ of smoothness s on a compact domain in $\mathbb{R}^d$, classical approximation theory gives $H(\varepsilon) \asymp \varepsilon^{-d/s}$. Solving the Yang–Barron balance equation:

$$\varepsilon_n^{-d/s} = n \varepsilon_n^2 \implies \varepsilon_n^{2+d/s} = n^{-1} \implies \varepsilon_n^2 \asymp n^{-\frac{2s}{2s+d}}. \quad (19)$$

Hence, the minimax excess risk scales as a power law with exponent:

$$\alpha_D = \frac{2s}{2s+d}. \quad (20)$$

3 Statistical Physics of Language and Hilberg’s Law

3.1 Hilberg’s Conjecture and Long-Range Context Correlations

In 1990, Wolfgang Hilberg [5] formulated a fundamental conjecture regarding the structural statistics of natural language: the mutual information $I(X_1^n; X_{n+1}^{2n})$ between two consecutive blocks of n words/characters does not decay exponentially (as in finite-state Markov chains), but scales as a sublinear power law:

$$I(n) \equiv I(X_1^n; X_{n+1}^{2n}) \propto n^\beta, \quad 0 < \beta < 1 \quad (\text{typically } \beta \approx 0.5). \quad (21)$$

Differentiating with respect to n implies that the conditional entropy of the $(n+1)$ -th token given the preceding n tokens converges to the infinite-context entropy floor H_∞ via a power law:

$$H(X_{n+1} \mid X_1^n) - H_\infty \sim n^{-\gamma}, \quad \text{where } \gamma = 1 - \beta. \quad (22)$$

3.2 Derivation of the Data Scaling Exponent: Cagnetta et al. (2024)

A breakthrough derivation by Cagnetta, Raventós, Ganguli, and Wyart [6] established an analytical link between Hilberg’s law, token autocorrelation statistics, and the empirical data scaling exponent α_D .

Theorem 3.1 (Cagnetta–Raventós–Ganguli–Wyart Theorem [6]). Assume that natural language satisfies:

1. **Conditional Entropy Scaling:** The optimal context-dependent excess entropy decays as:

$$H_n - H_\infty \sim n^{-\gamma}. \quad (23)$$

2. **Token Correlation Decay:** The two-point connected correlation function between tokens separated by distance n decays with exponent β :

$$|C(n)| \equiv |\mathbb{E}[f(X_t)g(X_{t+n})] - \mathbb{E}[f(X_t)]\mathbb{E}[g(X_{t+n})]| \sim n^{-\beta}. \quad (24)$$

When training on a finite dataset of P total tokens, empirical estimation of statistics at distance n is corrupted by finite-sample noise $\sigma_{\text{sample}} \sim \mathcal{O}(P^{-1/2})$. The maximum context distance $n^*(P)$ that a model can statistically resolve is determined by the signal-to-noise threshold:

$$|C(n^*(P))| \sim \frac{1}{\sqrt{P}} \implies (n^*(P))^{-\beta} \sim P^{-1/2} \implies \boxed{n^*(P) \sim P^{\frac{1}{2\beta}}}. \quad (25)$$

Substituting this maximum resolvable context length $n^*(P)$ into the conditional entropy decay law gives the excess empirical cross-entropy loss:

$$\mathcal{L}(P) - H_\infty \sim [n^*(P)]^{-\gamma} \sim \left(P^{\frac{1}{2\beta}}\right)^{-\gamma} = P^{-\frac{\gamma}{2\beta}}. \quad (26)$$

Therefore, the data scaling exponent is precisely:

$$\boxed{\alpha_D = \frac{\gamma}{2\beta}}. \quad (27)$$

Remark 3.1 (Self-Consistency and Empirical Agreement). Under Hilberg’s hypothesis, the mutual information scaling exponent β and the conditional entropy exponent γ are coupled via $\gamma = 1 - \beta$. If $\beta \approx 0.5$, then $\gamma \approx 0.5$, which predicts:

$$\alpha_D = \frac{0.5}{2(0.5)} = 0.50. \quad (28)$$

When tested empirically across synthetic text, WikiText-103, and TinyStories on GPT-2 and LLaMA architectures, Cagnetta et al. measured $\beta \approx 0.55\text{--}0.65$ and observed empirical data exponents $\alpha_D \in [0.35, 0.48]$, exhibiting close agreement between language statistics and neural loss scaling.

3.3 Information-Theoretic Neural Scaling: Jeon & Van Roy (2024)

Jeon and Van Roy (2024) [7] formalized neural scaling laws from rate–distortion and Bayesian information-theoretic principles. For an infinite-width two-layer network data-generating process with epistemic latent representation Z , they prove that the rate of information acquisition $\mathcal{I}(W; \mathcal{D}_P)$ between model weights W and dataset $\mathcal{D}_P$ dictates generalization:

$$\mathbb{E}[\mathcal{L}(\hat{\theta}) - \mathcal{L}^*] \leq \sqrt{\frac{2\mathcal{I}(W; \mathcal{D}_P)}{P}}. \quad (29)$$

By optimizing the model parameter rate $R(N)$ alongside dataset rate $R(P)$, they derive the near-linear optimal compute allocation frontier $N \propto P$ up to logarithmic factors, providing an information-theoretic underpinning for the empirical Chinchilla scaling rule [2].

4 Approximation Theory and Differential Geometry

4.1 Manifold Intrinsic Dimension: Sharma & Kaplan (2020)

High-dimensional data typically concentrates on a lower-dimensional Riemannian sub-manifold $\mathcal{M} \subset \mathbb{R}^{D_{\text{amb}}}$ of intrinsic dimension $d \ll D_{\text{amb}}$. Sharma and Kaplan (2020) [8] developed a geometric derivation of the parameter scaling exponent α_N based on piecewise-linear approximation on $\mathcal{M}$.

Theorem 4.1 (Manifold Tiling and Geometric Parameter Scaling). Let $f^* : \mathcal{M} \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ -smooth target function on a compact d -dimensional Riemannian manifold $\mathcal{M}$. A neural

network with N parameters can partition $\mathcal{M}$ into $K \asymp N$ local cells (Voronoi-like polyhedra) of characteristic linear size:

$$\varepsilon \asymp N^{-1/d}. \quad (30)$$

Within each cell centered at $x_0 \in \mathcal{M}$, the network fits a local affine approximation $\hat{f}(x) = f^*(x_0) + \nabla_{\mathcal{M}} f^*(x_0)^T (x - x_0)$. The local approximation error is governed by the second-order Taylor remainder:

$$f^*(x) - \hat{f}(x) = \frac{1}{2} (x - x_0)^T \nabla_{\mathcal{M}}^2 f^*(x_0) (x - x_0) + \mathcal{O}(\|x - x_0\|^3). \quad (31)$$

Integrating the squared error over $\mathcal{M}$ yields:

$$L(N) = \int_{\mathcal{M}} |f^*(x) - \hat{f}(x)|^2 d\mu(x) \asymp \varepsilon^4 \asymp (N^{-1/d})^4 = \boxed{N^{-4/d}}. \quad (32)$$

Hence, under second-order smooth regression on a manifold of intrinsic dimension d :

$$\alpha_N \approx \frac{4}{d}. \quad (33)$$

Remark 4.1 (First-Order vs. Second-Order Regression). If the target function is only Lipschitz continuous ($\mathcal{C}^1$) or if the network is restricted to piecewise-constant approximations, the error scales as $\varepsilon^2 \asymp N^{-2/d}$. For natural language, estimates of the intrinsic dimension of token representations yield $d \approx 10$ to 30 , which predicts $\alpha_N \in [0.13, 0.40]$, matching empirical observations.

4.2 Resolution-Limited vs. Variance-Limited Regimes: Bahri et al. (PNAS 2024)

Bahri, Dyer, Kaplan, Lee, and Sharma (PNAS 2024) [9] synthesized manifold geometry and statistical learning to identify two distinct, interacting scaling regimes.

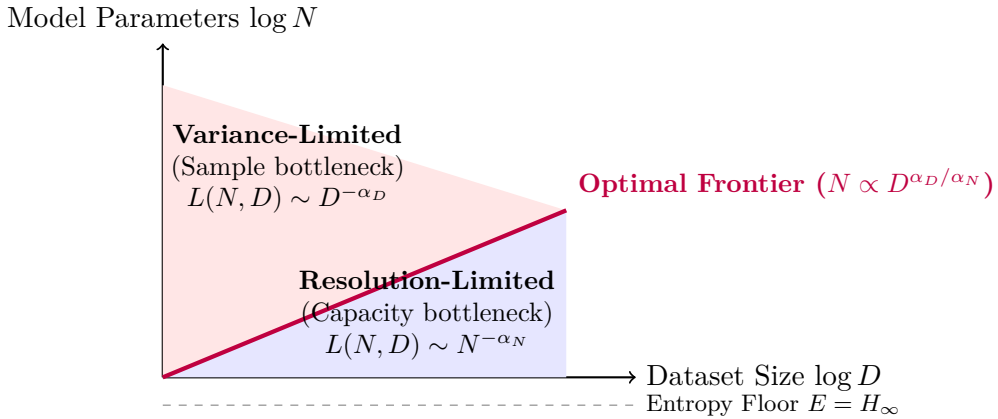


Figure 1: The two-regime scaling phase diagram of Bahri et al. [9]. The parameter-limited regime is governed by manifold resolution capacity, while the data-limited regime is governed by statistical sampling variance.

Definition 4.1 (Two Fundamental Regimes). 1. **Variance-Limited Regime** ($N \gg D$): The model has excess capacity to interpolate training samples. Generalization error is dominated by finite-sample estimation variance, scaling via Central Limit Theorem (CLT) effects as $L(D) \sim D^{-1}$ (parametric variance) or $D^{-4/d}$ (non-parametric sample resolution on the manifold).

2. **Resolution-Limited Regime** ($D \gg N$): Training data is effectively infinite. The model parameter capacity N limits the finest spatial/spectral frequency the network can resolve on $\mathcal{M}$, yielding $L(N) \sim N^{-4/d}$.

Bahri et al. show that the joint loss is characterized by a smooth harmonic crossover:

$$L(N, D) = E + \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\kappa}} + \left(\frac{D_c}{D} \right)^{\frac{\alpha_D}{\kappa}} \right]^\kappa, \quad (34)$$

where $\kappa > 0$ controls the sharpness of the transition between the resolution-limited and variance-limited regimes.

4.3 Fundamental Impossibility Lower Bounds in Approximation Theory

Theoretical scaling laws cannot exceed the fundamental mathematical limits of continuous approximation and parametric estimation. We catalog three foundational impossibility theorems.

Theorem 4.2 (DeVore–Howard–Micchelli Continuous Nonlinear n -Width Lower Bound (1989) [10]).

Let $W_p^s([0, 1]^d)$ denote the Sobolev ball of smoothness s in d dimensions. Let $\mathcal{M}_N = \{g(\theta, \cdot) : \theta \in \mathbb{R}^N\}$ be any parametric manifold of approximating functions generated by a continuous parameter selection map $\theta : W_p^s \rightarrow \mathbb{R}^N$. Then the continuous non-linear n -width satisfies:

$$d_N(W_p^s([0, 1]^d))_q \geq c \cdot N^{-s/d}. \quad (35)$$

This theorem proves that **no continuous parameter optimization method** (such as gradient descent on neural networks) can ever beat the curse of dimensionality $N^{-s/d}$ for worst-case functions in Sobolev spaces.

Theorem 4.3 (Rissanen Minimum Description Length (MDL) Redundancy Bound (1986, 1996) [11]). Let $\mathcal{M} = \{p_\theta : \theta \in \Theta \subset \mathbb{R}^k\}$ be a k -dimensional identifiable parametric model family with non-singular Fisher information matrix $I(\theta)$. For any probability model $q(X^n)$, the minimax parameter redundancy over n samples satisfies:

$$\inf_q \sup_{\theta \in \Theta} \mathbb{E}_{p_\theta} \left[\log \frac{p_\theta(X^n)}{q(X^n)} \right] \geq \frac{k}{2} \log n + \frac{k}{2} \log \left(\frac{1}{2\pi e} \right) + \log \int_\Theta \sqrt{\det I(\theta)} d\theta + o(1). \quad (36)$$

Rissanen’s theorem establishes that learning k parameters inevitably incurs a fundamental statistical cost of $\frac{k}{2} \log n$ bits of code length.

Theorem 4.4 (Statistical Query (SQ) and Cryptographic Hardness (Kearns 1998, Shamir 2020) [12, 13]). For the class of parity functions over d binary bits, any algorithm with access only to Statistical Queries with tolerance $\tau \geq 2^{-d/3}$ requires at least $2^{\Omega(d)}$ queries. Furthermore, under standard cryptographic assumptions (existence of one-way functions), there exist target functions for which no polynomial-size neural network can achieve loss advantage $> 2^{-\Omega(d)}$.

Implication of Impossibility Results

Neural scaling laws $L \sim N^{-\alpha}$ are **not universal mathematical laws of arbitrary functions**. They are contingent upon the underlying data distribution possessing specific structural properties: low intrinsic manifold dimension d , heavy-tailed spectral decay, and hierarchical compositional structure.

5 Statistical Physics, Kernel Theory, and Spectral Decay

5.1 Exact Learning Curves from Spectral Decomposition: Bordelon et al. (2020)

In the infinite-width Neural Tangent Kernel (NTK) or Random Feature regime, neural network generalization can be solved exactly using statistical mechanics and replica theory. Bordelon, Canatar, and Pehlevan (2020) [14] and Canatar et al. (2021) [15] established the analytical mapping from covariance spectra to generalization error power laws.

Theorem 5.1 (Spectral Decay to Generalization Error). Let the kernel integral operator $T_K f(x) = \int K(x, x') f(x') d\mu(x')$ admit an eigendecomposition with eigenvalues λ_k and orthonormal eigenfunctions $\psi_k(x)$:

$$K(x, x') = \sum_{k=1}^{\infty} \lambda_k \psi_k(x) \psi_k(x'). \quad (37)$$

Suppose kernel eigenvalues decay as a power law of degree $a > 1$, and target function $f^*(x) = \sum_k c_k \psi_k(x)$ has projection coefficients decaying as $b > 1$:

$$\lambda_k \asymp k^{-a}, \quad c_k^2 \asymp k^{-b}. \quad (38)$$

Then, when trained on D independent samples with ridge regularization $\lambda \rightarrow 0^+$, the asymptotic generalization error $\mathcal{E}(D) = \mathbb{E}_x[(f^*(x) - \hat{f}_D(x))^2]$ satisfies:

$$\boxed{\mathcal{E}(D) \asymp D^{-\frac{a+b-1}{a}}}. \quad (39)$$

Corollary 5.2 (Target-Prior Alignment). When the target function is drawn from the Gaussian process prior associated with the kernel covariance ($c_k^2 \sim \lambda_k \implies b = a$), the generalization error exponent simplifies to:

$$\alpha_D = \frac{2a-1}{a} = 2 - \frac{1}{a}. \quad (40)$$

As the kernel spectrum becomes steeper ($a \rightarrow \infty$), the data scaling exponent approaches the classical parametric rate $\alpha_D \rightarrow 2$ (under MSE loss) or $\alpha_D \rightarrow 1$ (under root MSE).

5.2 Rigorous Joint (M, N) Linear Scaling Laws: Lin et al. (NeurIPS 2024)

Lin, Dobriban, and Hong (NeurIPS 2024) [16] established exact asymptotic and non-asymptotic scaling bounds for high-dimensional ridge regression under simultaneously varying feature dimension M (parameter count) and sample size N (dataset size).

Theorem 5.3 (Lin, Dobriban, Hong (2024) [16]). Consider high-dimensional linear regression with data covariance Σ having power-law spectral decay $\lambda_k(\Sigma) \asymp k^{-a}$ with $a > 1$. The excess prediction error $\mathcal{E}(M, N)$ with M features and N samples satisfies:

$$\boxed{\mathcal{E}(M, N) = \Theta\left(M^{-(a-1)} + N^{-\frac{a-1}{a}}\right)}. \quad (41)$$

This result rigorously proves that parameter exponent $\alpha_M = a - 1$ and data exponent $\alpha_N = (a - 1)/a$ are linked to the spectral exponent a of the data distribution, providing an exact mathematical verification of the spectral scaling hypothesis.

6 Combinatorial Quanta and Discrete Skill Models

6.1 The Quantization Model of Neural Scaling: Michaud et al. (NeurIPS 2023)

Michaud et al. (2023) [18] proposed an alternative combinatorial model of neural scaling: the **Quantization Model**. Rather than modeling data as a continuous manifold or Gaussian process, knowledge is conceptualized as a collection of discrete sub-problems, facts, or skills (“quanta”) $\mathcal{Q} = \{q_1, q_2, \dots\}$.

Definition 6.1 (Quantum Structure and Utility). Each quantum q_i is characterized by:

1. **Utility / Frequency (U_i):** The loss reduction achieved when quantum q_i is mastered. The utilities follow a discrete power law (Zipf’s law):

$$U_i = c_U \cdot i^{-\beta_Q}, \quad \beta_Q > 1. \quad (42)$$

2. **Capacity Cost (C_i):** The parameter count required to store and execute the circuit for quantum q_i , scaling as $C_i = c_C \cdot i^\nu$.

Theorem 6.1 (Quantization Loss Scaling). Under optimal greedy learning dynamics, a model with N total parameters allocates capacity to the top $K(N)$ most utility-dense quanta. If capacity cost is uniform across quanta ($\nu = 0$, $C_i \approx 1$), the number of learned quanta is $K(N) \asymp N$. The residual excess loss from unlearned quanta is:

$$L(N) - E = \sum_{i=K(N)+1}^{\infty} U_i \approx \int_N^{\infty} c_U x^{-\beta_Q} dx = \frac{c_U}{\beta_Q - 1} N^{-(\beta_Q-1)}. \quad (43)$$

Therefore, the parameter scaling exponent is directly dictated by the Zipfian utility exponent:

$$\boxed{\alpha_N = \beta_Q - 1}. \quad (44)$$

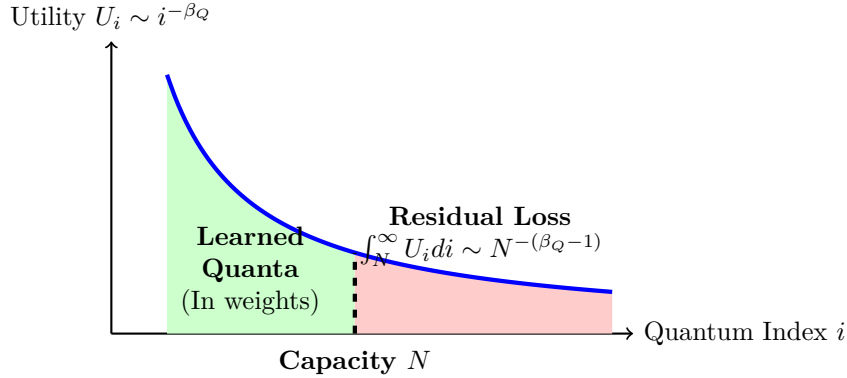


Figure 2: The Quantization Model of Neural Scaling (Michaud et al. [18]). Network parameters absorb high-frequency Zipfian skills first; remaining tail quanta determine the power-law loss decay.

6.2 Universal Algorithmic Priors: Hutter (2021)

Marcus Hutter (2021) [19] derived scaling laws from algorithmic information theory and Kolmogorov complexity. Under Levin’s universal algorithmic probability $M(x) = \sum_{p: U(p)=x} 2^{-\ell(p)}$, programs are distributed according to power laws in length and execution time. Learning algorithms that sequentially discover algorithmic regularities experience power-law reduction in regret, bridging universal compression with empirical deep learning scaling.

7 Representation Geometry, Superposition, and Capacity

7.1 Toy Models of Superposition: Elhage et al. (Anthropic 2022)

A central question in deep representation learning is how a network with activation dimension d_{model} can represent $M \gg d_{\text{model}}$ distinct semantic features. Elhage et al. [20] formalized this phenomenon as **neural superposition**.

Theorem 7.1 (Almost-Orthogonality and the Johnson–Lindenstrauss Geometry). Let $\mathcal{S} = \{x_1, \dots, x_M\} \subset \mathbb{R}^M$ be sparse features with activation probability $p \ll 1$. Let $\mathbf{W} \in \mathbb{R}^{d_{\text{model}} \times M}$ be a feature embedding dictionary with unit-norm columns $\|w_i\|_2 = 1$. By the Johnson–Lindenstrauss Lemma, for any $\varepsilon \in (0, 1)$, if:

$$d_{\text{model}} \geq \frac{8 \ln M}{\varepsilon^2}, \quad (45)$$

there exists a dictionary $\mathbf{W}$ such that all off-diagonal cross-talk interferences satisfy:

$$|\langle w_i, w_j \rangle| \leq \varepsilon \quad \forall i \neq j. \quad (46)$$

Proposition 7.2 (Cross-Talk Interference Noise Scaling). When reconstructing feature x_i from compressed activation vector $h = \mathbf{W}x$, reconstruction with linear readout yields:

$$\hat{x}_i = w_i^T h = x_i + \sum_{j \neq i} \langle w_i, w_j \rangle x_j. \quad (47)$$

The cross-talk interference variance is:

$$\mathbb{E} \left[\left(\sum_{j \neq i} \langle w_i, w_j \rangle x_j \right)^2 \right] = p(1-p) \sum_{j \neq i} |\langle w_i, w_j \rangle|^2 \asymp \frac{pM}{d_{\text{model}}}. \quad (48)$$

As the model width d_{model} scales, the interference noise decays as $\mathcal{O}(1/d_{\text{model}})$, yielding a power-law error decay in representation capacity until the network transitions from the superposed regime ($M > d_{\text{model}}$) to the orthogonal regime ($M \leq d_{\text{model}}$).

8 Comparative Synthesis and Taxonomy of Theories

To clarify how these theoretical perspectives relate, Table 1 synthesizes the assumptions, mathematical machinery, derived exponents, and domain applicability across all major scaling theories.

9 The “Honest Gap” and Open Theoretical Challenges

While the theoretical landscape has achieved major mathematical breakthroughs, a critical foundational gap persists between theoretical derivations and empirical practice:

The Fundamental “Honest Gap”

No existing theory derives the scaling exponents α_N, α_D for real-world natural language purely from first physical principles. Existing theories succeed in proving that the scaling exponents are rigorous functions of underlying data statistics:

$$\alpha = \Phi(\beta_{\text{spectral}}, d_{\text{intrinsic}}, \beta_{\text{Zipf}}, \gamma_{\text{entropy}}, \beta_{\text{correlation}}). \quad (49)$$

However, these data statistics $(a, d, \beta, \gamma, \beta_Q)$ are **empirically measured rather than mathematically derived**. The ultimate challenge of deep learning theory is to derive why natural human language and physical multimodal data exhibit the specific spectral exponents and intrinsic dimensions that they do.

Theoretical Framework	Core Assumptions	Mathematical Machinery	Derived Scaling Exponent	Key References
Rate-Distortion Theory	Heavy-tailed Gaussian source, $\lambda_k \sim k^{-(1+\beta)}$	Shannon Rate-Distortion, Reverse Water-Filling	$D(R) \sim R^{-\beta}$, $L(N) \sim N^{-\beta}$	Shannon (1951), Bordelon (2020)
Metric Entropy Minimax	Target class in Sobolev / Besov space $H(\varepsilon) \sim \varepsilon^{-d/s}$	Yang-Barron Minimax, Kolmogorov ε -Entropy	$\alpha_D = \frac{2s}{2s+d}$	Yang & Barron (1999) [4]
Language Statistics & Hilberg	Text correlation $ C(n) \sim n^{-\beta}$, entropy decay $H_n - H_\infty \sim n^{-\gamma}$	Finite-sample estimation noise cutoff $n^*(P) \sim P^{\frac{1}{2\beta}}$	$\alpha_D = \frac{\gamma}{2\beta} \approx \frac{1-\beta}{2\beta}$	Cagnetta et al. (2024) [6], Hilberg (1990) [5]
Manifold Tiling Geometry	Data on Riemannian manifold $\mathcal{M}$ of intrinsic dimension d	Local Voronoi partition, 2nd-order Taylor remainder	$\alpha_N = \frac{4}{d}$ (or $\frac{2}{d}$)	Sharma & Kaplan (2020) [8]
Resolution vs. Variance Phase Diagram	Interplay of finite parameter count N and finite samples D	Non-parametric manifold resolution + CLT variance	$L(N, D) \sim N^{-4/d} + D^{-1}$	Bahri et al. (PNAS 2024) [9]
Kernel Spectra & NTK Physics	Power-law kernel spectrum $\lambda_k \sim k^{-a}$, target decay $c_k^2 \sim k^{-b}$	Replica method, Random Matrix Theory, NTK	$\alpha_D = \frac{a+b-1}{a}$	Bordelon et al. (2020) [14], Lin et al. (2024) [16]
Quantization / Discrete Skills	Independent skills/quanta with Zipfian utilities $U_i \sim i^{-\beta_Q}$	Greedy capacity allocation, tail integration	$\alpha_N = \beta_Q - 1$	Michaud et al. (2023) [18]
Feature Superposition	$M \gg d_{\text{model}}$ sparse features, almost-orthogonal frames	Johnson-Lindenstrauss Lemma, cross-talk variance	Noise $\propto \frac{pM}{d_{\text{model}}}$	Elhage et al. (Anthropic 2022) [20]

Table 1: Comprehensive taxonomy of theoretical frameworks for neural scaling laws.

9.1 Key Open Frontiers

1. **Breakdown of Power Laws and Non-Asymptotic Regimes:** Scaling laws break down in specific regimes:
 - *Sub-power-law saturation:* When approaching irreducible entropy floor $E = H_\infty$.
 - *Emergent abilities and Grokking:* Discontinuous phase transitions in discrete multi-step reasoning tasks (e.g., modular arithmetic, in-context proof search) that do not exhibit smooth power-law progress on macroscopic task accuracy.
2. **Autoregressive Architecture Specifics:** Most kernel and manifold theories treat neural networks as general continuous function approximators. Incorporating causal multi-head self-attention mechanisms, rotary position embeddings, and softmax routing into spectral scaling theories remains an active open direction.
3. **Compute-Optimal Inference and Test-Time Search Scaling:** Recent frontiers (e.g., OpenAI o1/o3) demonstrate that scaling test-time compute (tree search, verification, Monte Carlo rollouts) yields power-law performance gains distinct from pretraining scaling. A unified rate-distortion theory bridging pretraining parameter rate with inference search budget represents the next major milestone.

10 Conclusion

Neural scaling laws are not arbitrary empirical accidents; they are the rigorous macroscopic manifestation of underlying microscopic mathematical structures.

By analyzing scaling through the lens of Shannon rate-distortion theory, we establish that power laws are the inevitable signature of heavy-tailed, infinite-dimensional data spectra under reverse water-filling. Through high-dimensional geometry and statistical physics, parameter and data exponents are mapped to the intrinsic manifold dimension, kernel eigenspectra, and long-range correlation decay of natural data distributions. While fundamental lower bounds (Yang-Barron, DeVore-Howard-Micchelli, Rissanen) set unbreakable speed limits on learning, the rich hierarchical structure of natural language permits modern deep neural networks to operate along an optimal information-theoretic frontier.

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